Data Flow Diagram (DFD)

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

Level 0: Context Data Flow & External Interfaces

Entity	Data Inflow to System	Data Outflow from System
Investigator	• Crime Scene Photos (JPG/PNG) • 911 Audio / Witness Tapes (WAV/MP3) • CCTV Surveillance Video (MP4) • Natural Language Questions	• Interactive 3D Spatial Crime Scene • Second-by-Second Synced Timeline • Grounded AI Answers & Summaries • Court-Admissible PDF Dossiers
Model Registry	• Model weights (RT-DETR, CLIP, Whisper) • Historical case sessions JSON	• Session context & timeline records • Generated PDF reports in /data/reports/
Groq Cloud API	• LLaMA-3.3 forensic hypotheses • Contextual answers & 4 translations	• Structured prompt payloads containing detected objects and timeline keypoints

Level 1: Functional Subsystem Data Flow Decomposition

Process	Input	Algorithm / Engine	Output Destination
P1: Vision	2D Scene Image	RT-DETR object detection + Depth estimation [X,Y,Z].	BBoxes, 3D Mesh -> SessionStore
P2: Audio	Audio Waveform	Whisper ASR Speech-to-Text + Groq translation.	Transcript, translations -> SessionStore
P3: Video	CCTV Video	1.0 Hz Frame sampler + BLIP captioning + Anomaly.	1Hz Timeline -> SessionStore
P4: Fusion	P1, P2, P3 data	Cross-modal timestamp synchronization & consistency.	Unified Dossier -> SessionStore
P5: PDF	Session JSON	ReportLab Platypus synthesis + XML character sanitation.	PDF File -> /data/reports/